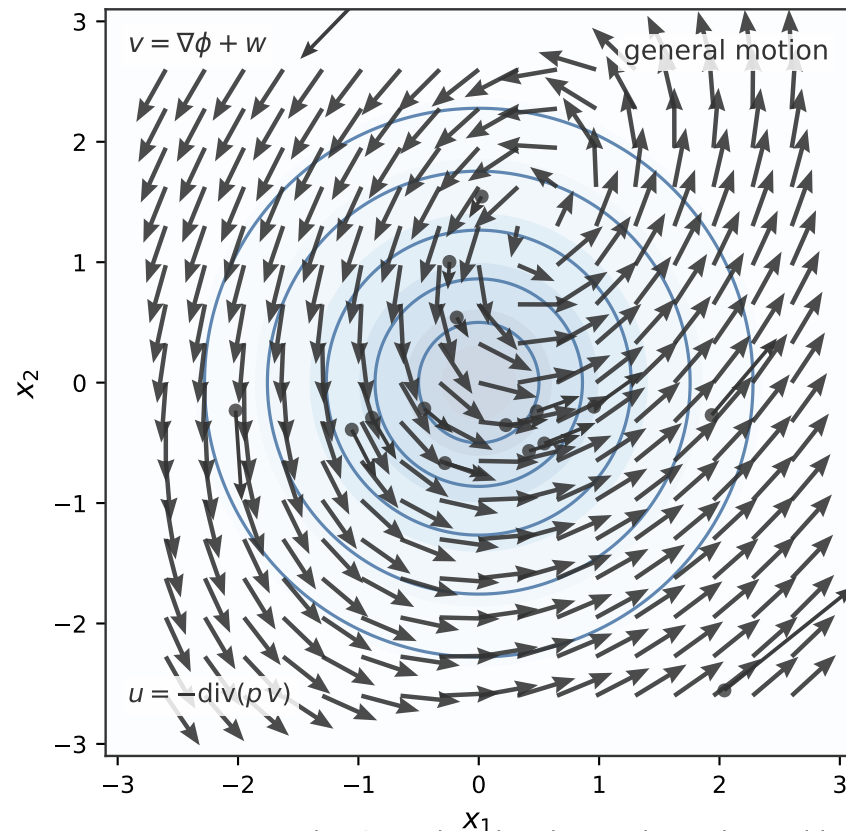


Weighted Helmholtz decomposition in Wasserstein-Otto geometry

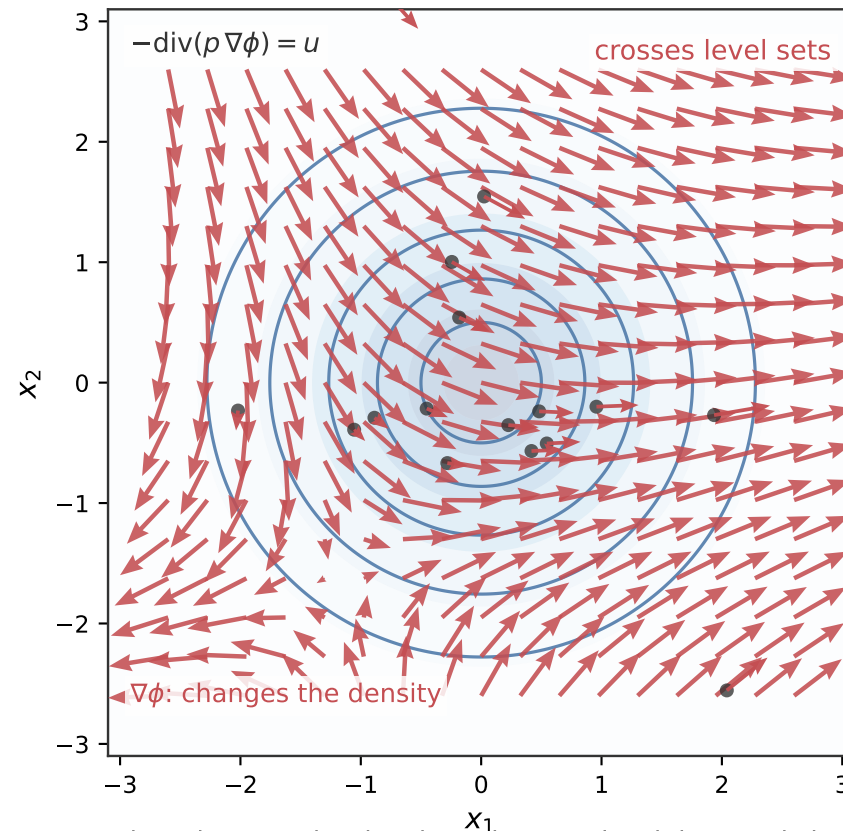
Arbitrary velocity field v



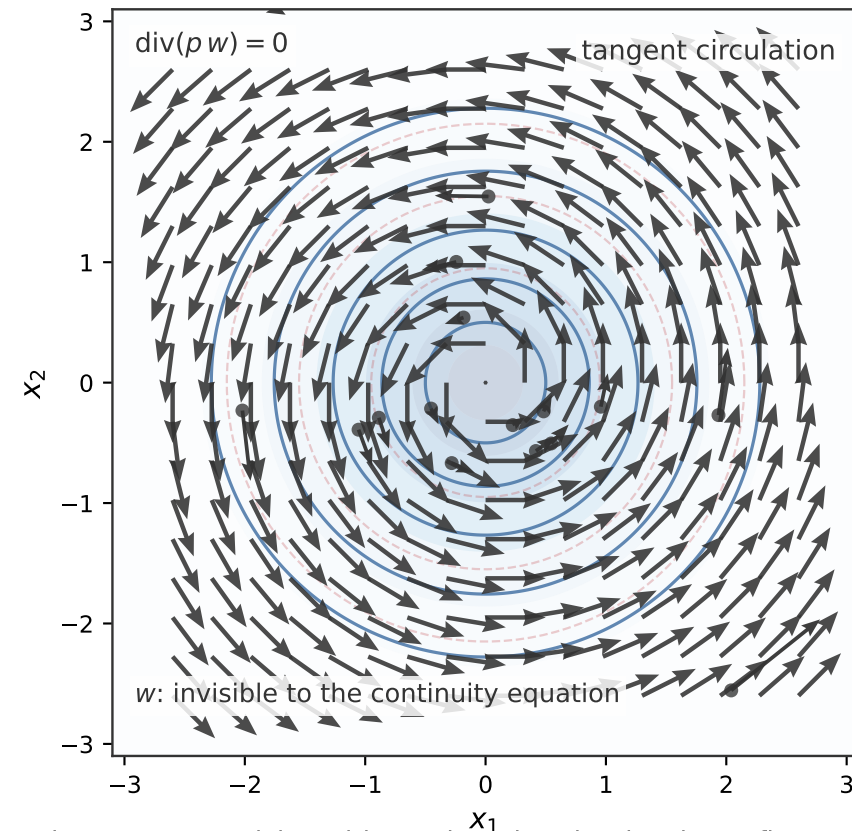
$$v = \nabla \phi + w, \quad \|v\|_p^2 = \|\nabla \phi\|_p^2 + \|w\|_p^2$$

$$\operatorname{div}(p w) = 0, \quad \Rightarrow \text{The initial representative}$$

$$-\operatorname{div}(p \nabla \phi) = u, \quad \text{Gradient component } \nabla \phi$$



Weighted divergence-free part w



Blue contours represent the Gaussian density p . The red panel isolates the component that changes the density, whereas the right panel shows a circulation that moves particles without changing the density at first order.